

PhD Project Outline

Value of Information for Investment Decisions under Uncertainty

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Target institution

ETH Zurich, Department of Computer Science (D-INFK) — Optimization & Decision Intelligence (ODI) Group, Institute for Machine Learning (Prof. Niao He). A domain co-advisor at HSG (private-markets / entrepreneurial finance) is envisaged.

1. Research context and focus

Investors in private markets decide under extreme uncertainty on fragmentary, non-standardised information. The dominant AI framing — *predict* which company will succeed — is increasingly crowded and, on its own, of limited practical value: an investor does not buy a prediction, they buy **information** that improves a decision. This project studies the prior, under-explored question: given a decision under uncertainty, *which* missing piece of information — at what cost — most improves the decision?

This builds on my accepted WI2026 Student Track paper (*Can Non-Financial Signals Price Private Companies?*), which shows on 3,403 PitchBook deals, under a strict out-of-time holdout, that ML on non-financial signals alone prices private companies competitively with financial baselines, and that **investor-syndicate structure** — not firm financials — is the dominant driver. That result exposes the ceiling of a purely predictive approach and motivates the shift from *prediction* to *value of information*. The natural home for this shift is decision-making under uncertainty: the acquisition of information is itself a sequential decision to be optimised, not a prediction to be improved.

2. Research questions

Main question: In investment decisions under uncertainty, which information acquisition — at what cost — most improves decision quality?

- How can each decision be represented by its *information state*: what is knowable at time t , reconstructed point-in-time, before the outcome is realised?
- How do we quantify the *marginal value* of acquiring an additional piece of information (a signal, a due-diligence step) relative to its cost?
- Can the acquisition process be framed as a sequential decision / optimisation-under-uncertainty problem with regret or optimality guarantees on *which* step to take next under a cost budget?

3. Theoretical framework

The project sits at the intersection of decision theory under uncertainty and information economics, operationalised with modern optimisation and ML. The core object is the *value of information*: the expected improvement in decision quality from moving $\text{Decision}(\text{state}) \rightarrow \text{acquire info (cost } c) \rightarrow \text{state}' \rightarrow \text{Decision}(\text{state}')$. The natural formal home is the ODI group's own language — the trade-off between the value of more information and its cost. This is precisely the object of the group's work on the *bias–variance–cost trade-off of stochastic optimisation* (NeurIPS 2021): acquiring more or better information reduces error but is costly, and the question is how much to acquire and when to stop. My project transports that trade-off from the estimation setting into a *decision* setting — where the "sample" is a diligence step and the "error" is decision regret. Two

further group results supply the sequential machinery: constrained-MDP formulations with regret guarantees (*Truly No-Regret Learning in Constrained MDPs*, ICML 2024) formalise "acquire the highest-value information subject to a cost/time budget," and stochastic optimisation under uncertainty for revenue management (*Operations Research*, 2024) is a template for turning a real allocation decision into a tractable objective. The methods exist; their application to private-market investment decisions is open.

4. Methodology

Feasibility does not depend on proprietary data. Ground truth comes from a *point-in-time reconstruction* of public sources (so the information state at time t excludes look-ahead) plus a *virtual-fund simulator* that lets us measure decision quality counterfactually — a natural sequential-decision environment in which an acquisition policy can be learned and evaluated. On top of this I formalise value-of-information estimators and an acquisition policy that selects the next diligence step by expected marginal value per unit cost, benchmarked against the WI2026 pipeline as a predictive baseline.

- **Year 1:** formalise the information-state / value-of-information objective; build the point-in-time dataset and simulator; first acquisition-policy experiments.
- **Year 2:** complete and submit the flagship paper (if it grows: a follow-up on how to evaluate whether a decision was good given what was knowable, not by its eventual outcome).
- **Year 3–4:** extensions (robustness, portfolio-level acquisition), writing, publications.

5. Expected contribution

The project contributes (i) a formalisation of value of information for private-market decisions as a sequential optimisation-under-uncertainty problem; (ii) a reproducible point-in-time and simulator methodology that makes the question testable without proprietary data; and (iii) empirical evidence on which information is worth acquiring in VC diligence. The contribution is domain-general: the same formulation applies beyond venture capital (private equity, credit, M&A), which is what makes it a research line rather than a single application.

6. Fit with host institution

The ODI group's stated focus — algorithmic and theoretical foundations for solving data-driven decision-making problems, large-scale optimisation, optimisation under uncertainty, and reinforcement learning — is precisely the toolbox this question needs. Framing value of information as a sequential / constrained decision problem is a direct extension of the group's optimisation-under-uncertainty and constrained-MDP work into a new, high-stakes application domain. The methodological core would live in your group; a private-markets co-advisor at HSG would anchor the domain, giving the "ML rigour + domain credibility" pairing. I note the group's SNSF Starting Grant and current openings, and would be glad to discuss whether this fits a funded position.

7. Selected references

- Howard, R. A. (1966). *Information Value Theory*. IEEE Trans. Systems Science and Cybernetics, 2(1), 22–26.
- Savage, L. J. (1954). *The Foundations of Statistics*. Wiley.
- Hu, Y., Chen, X., & He, N. (2021). *On the Bias-Variance-Cost Tradeoff of Stochastic Optimization*. NeurIPS 2021.

- Müller, A., Alatur, P., Cevher, V., Ramponi, G., & He, N. (2024). *Truly No-Regret Learning in Constrained MDPs*. ICML 2024.
- Chen, X., He, N., Hu, Y., & Ye, Z. (2024). *Efficient Algorithms for a Class of Stochastic Hidden Convex Optimization and Its Applications in Network Revenue Management*. Operations Research.
- Guidi, A., Rashid, S., & Zhong, H. (2026). *Can Non-Financial Signals Price Private Companies?* WI2026 Student Track (accepted).

One-page WI2026 summary attached. Code and manuscript available on request.